

Quotient rule



- 1 To differentiate $y = \frac{3x + 4}{4x - 3}$ using the quotient rule, let $u = 3x + 4$ and $v = 4x - 3$, then:
 - a Find u' .
 - b Find v' .
 - c Hence, find y' .
 - d Is it possible for the derivative of this function to be zero?

- 2 To differentiate $y = \frac{5x}{8x - 9}$ using the quotient rule, let $u = 5x$ and $v = 8x - 9$, then:
 - a Find u' .
 - b Find v' .
 - c Hence, find y' .
 - d Is it possible for the derivative of this function to be zero?

- 3 To differentiate $y = \frac{9x^2}{2x + 8}$ using the quotient rule, let $u = 9x^2$ and $v = 2x + 8$, then:
 - a Find u' .
 - b Find v' .
 - c Hence, find y' .
 - d Is it possible for the derivative of this function to be zero?

- 4 To differentiate $y = \frac{4x^2 + 3}{2x^2 + 5}$ using the quotient rule, let $u = 4x^2 + 3$ and $v = 2x^2 + 5$, then:
 - a Find u' .
 - b Find v' .
 - c Hence, find y' .
 - d Is it possible for the derivative of this function to be zero?

- 5 Differentiate each of the following:
 - a $y = \frac{4x + 3}{4x - 3}$
 - b $y = \frac{4x - 3}{3x - 4}$
 - c $y = \frac{2x}{4x^2 - 3}$
 - d $y = \frac{5x + 3}{4x^2 - 1}$
 - e $y = \frac{2 - x}{7x + 3}$
 - f $y = \frac{x^2}{4x^3 + 1}$
 - g $y = \frac{3x + 2}{6 + 5x^2}$
 - h $y = \frac{\sqrt{x} + 2}{9\sqrt{x} + 8}$

- 6 Find the gradient function of $y = \frac{x^2 - 5x - 2}{x + 8}$.



7 Consider the function $y = \frac{3}{x}$.

- a By first rewriting it in negative index form, differentiate y .
- b By using the substitutions $u = 3$ and $v = x$, differentiate y using the quotient rule.
- c Find the value of x for which the gradient is undefined.

8 Consider the function $y = \frac{1}{8x + 3}$.

- a Differentiate the function using the chain rule.
- b Differentiate the function using the quotient rule.

9 Consider the function $y = \frac{3 - 4x}{3x - 4}$.

- a Differentiate y .
- b Is it possible for the derivative to be zero?
- c Find the value of x that will make the derivative undefined.

10 For each of the following functions:

- i Differentiate y .
- ii Is it possible for the derivative to be zero?
- iii Find the value of x that will make the derivative undefined.
- iv State the region(s) of the domain for which the function is increasing.
- v State the region(s) of the domain for which the function is decreasing.

a $y = \frac{x + 6}{x - 6}$

b $y = \frac{5x}{3x - 4}$

c $y = \frac{4x}{5 - 2x}$

d $y = \frac{2x + 3}{2x - 3}$

11 Differentiate each of the following:

a $y = \frac{(4t^2 + 3)^3}{(5 + 2t)^5}$

b $y = \sqrt{\frac{2 + 7x}{2 - 7x}}$



Gradients and tangents

- 12 Find $f'(4)$ for $f(x) = \frac{5x}{16 + x^2}$.
- 13 Find $f'(1)$ for $f(x) = \frac{8}{2 + 2x^2}$.
- 14 Find the value of $f'(0)$ if $f(x) = \frac{x}{\sqrt{16 - x^2}}$.
- 15 Consider the function $f(x) = \frac{4x^9}{(x + 2)^4}$.
- a Find $f'(2)$.
- b Is the function increasing or decreasing at $x = 2$?
- 16 Find the gradient of $f(x) = \frac{(x + 7)^9}{(x + 9)^4}$ at $x = -5$.
- 17 Find the gradient of the tangent to the curve $y = \frac{9x}{4x + 1}$ at the point $\left(1, \frac{9}{5}\right)$.
- 18 Find the equation of the tangent to $y = \frac{x}{x + 4}$ at the point $\left(8, \frac{2}{3}\right)$.
- 19 Find the equation of the tangent to $y = \frac{x^2 - 1}{x + 3}$ at the point where $x = 4$.
- 20 Consider the function $g(x)$ defined as $g(x) = \frac{f(x)}{x^3 + 3}$, where $f(x)$ is a function of x .
Given that $f(2) = 5$ and $f'(2) = 6$, determine the value of $g'(2)$.
- 21 Find the values of x such that the gradient of the tangent to the curve $y = \frac{6x - 1}{3x - 1}$ is -3 .
- 22 Differentiate $y = \frac{x^2}{x + 3}$ and find the value of a if $y' = 0$ at $x = a$.
- 23 Differentiate $y = \frac{x^2 + k}{x^2 - k}$ and find the possible values of k given that $y' = 1$ at $x = -3$.